

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Six Semester of B.Tech. (IT) Examination

June 2022

IT378 WIRELESS COMMUNICATION & MOBILE COMPUTING**Date: 30/06/2022, Thursday Time: 10:00 a.m. to 01:00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION – I**Q - 1 Do as directed.**

- (a) Which of the following spread spectrum techniques were used in the original IEEE 802.11 standard? [01]
- a. FHSS and DSSS
 - b. THSS and FHSS
 - c. THSS and DSSS
 - d. Hybrid technique
- (b) HIPER-LAN stands for _____. [01]
- a. High Precision Radio Local Area Network
 - b. High Performance Radio Local Area Network
 - c. High Precision Radio Land Area Network
 - d. Huge Performance Radio Link Access Node
- (c) Mobile station MS is combination of ____ + _____. [01]
- a. Mobile Equipment (ME) , International Mobile Subscriber Identity (IMSI)
 - b. Mobile Equipment (ME), personal identification code (PIN)
 - c. Subscriber Identity Module (SIM), Mobile Equipment (ME)
 - d. All of above.
- (d) GSM use the Combination _____ and _____ multiple access. [01]
- a. CDMA, FDMA
 - b. TDMA, CDAM
 - c. FDMA, TDMA
 - d. CDMA, SDMA
- (e) Cells under the control of different BSCs but belonging to the same Mobile Switching Center (MSC) is called [01]
- a. Inter-MSC handover
 - b. Inter-cell, intra-BSC handover
 - c. Intra-cell handover
 - d. Inter-BSC, Intra- MSC handover

Q - 2 (a) Which information are stored in Home Location Register (HLR) in GSM. [05]

OR

Q - 2 (a) Draw process diagram Mobile Terminated Call (MTC) and Mobile Originated Call (MOC). [05]

Q - 2 (b) What is signal? Explain signal Propagation ranges with figure. [04]

Q - 2 (c) List out the types of multiplexing. Explain each in details with diagram. [06]

OR

Q - 2 (c) Explain GSM architecture with proper diagram. [06]

Q - 3 (a) Explain the following terms. [06]

1. Fading
2. Scattering
3. EIR
4. PLMN
5. MSISDN
6. Cell

Q - 3 (b) Explain GPRS with diagram. [05]

OR

Q - 3 (b) Write a short note on DECT. [05]

Q - 3 (c) Explain UMTS hierarchical cell structure. [04]

SECTION - II

Q - 4 Do as directed

(a) _____ segments a TCP connection into a fixed part and wireless part. [01]

- a. I-TCP
- b. M-TCP
- c. Both
- d. None of the above

(b) LEO is having lifetime of _____ years. [01]

- a. 5-8
- b. 8-10
- c. 10-12
- d. None of the above

(c) Indirect TCP does not propagate transmission errors on the wireless link into a fixed network [01]

- True
- False

(d) Encryption schemes fails snooping TCP concept? State true or false [01]

- True
- False

- (e) The scatternet is a combination of _____ [01]
- Multiple piconet
 - Double Piconet
 - Single piconet
 - None of the above
- (f) Which of the following buffers data close to the mobile host to perform fast local retransmission in case of packet loss. [01]
- Snooping TCP
 - I-TCP
 - Foreign Agent
 - None of the above
- (g) The behavior of TCP shows after the detection of congestion is called [01]
- Slow Down
 - Wait to restart
 - Slow start
 - None of the above

Q - 5 Answer any four questions.

- (a) Explain how data transfer to and from the mobile system using diagram and discuss below components with respect to diagram. [07]
- Mobile Node
 - Home Agent
 - Foreign Agent
 - Care-of Address
 - Correspondent Node
- (b) What is Mobile TCP? Discuss Indirect TCP and Snooping TCP. [07]
- (c) What is WLAN? List out the WLAN technologies. Explain one in detail. [07]
- (d) Explain Bluetooth in details. [07]
- (e) Explain various encapsulation techniques in mobile IP. [07]
- (f) Compare traditional routing algorithm with mobile routing algorithm (DSDV). [07]
